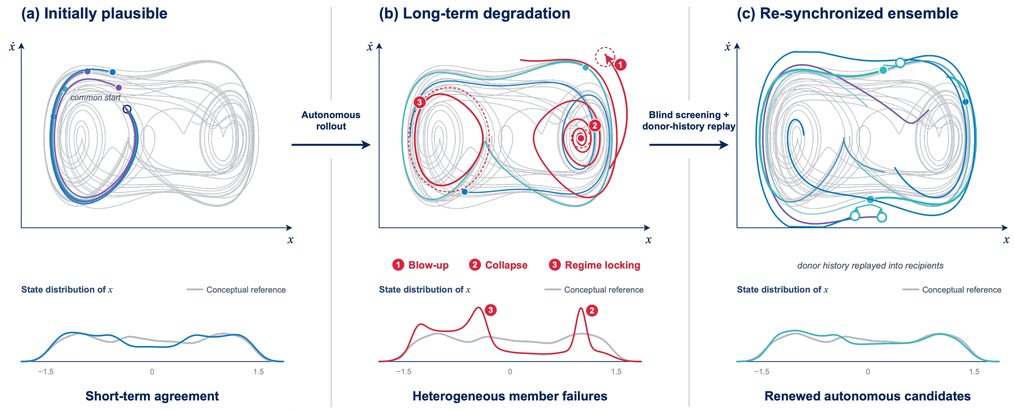
# Resynchronization Keeps Reservoir Ensembles Healthy for Long-Term Prediction

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## Motivation

Ensemble methods combine multiple models to obtain more reliable predictions. Yet maintaining a useful ensemble over long autonomous rollouts is challenging: individual members may drift into unrealistic dynamics, even when they were initially well trained.

Reservoir Computing (RC) offers an efficient framework for building such ensembles. Beyond the pathwise predictability horizon, however, reservoir members may exhibit blow-up, fixed-point collapse, or persistent regime locking, compromising long-term statistical prediction.




*The challenge is not simply to discard failing members, but to recover them while preserving a healthy and diverse ensemble without access to future observations.*

## Research Question

Can blind resynchronization sustain a healthy and diverse reservoir ensemble beyond the pathwise predictability horizon?

**Q1: Ensemble Healthiness**
Can blind resynchronization maintain dynamically plausible reservoir trajectories that reproduce the target system's long-term statistical properties?

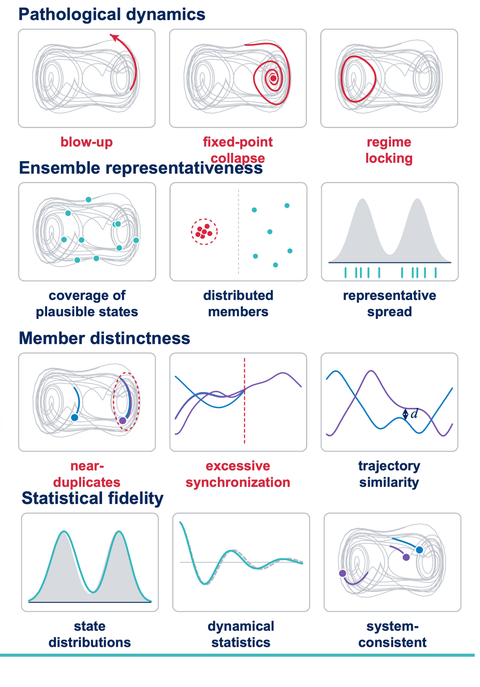


A healthy ensemble requires both dynamical validity and statistical fidelity.

**Q2: Ensemble Diversity**
Can ensemble healthiness be maintained without excessive synchronization or loss of diversity among members?



Diversity requires distinct, non-redundant ensemble members.




## Methodology

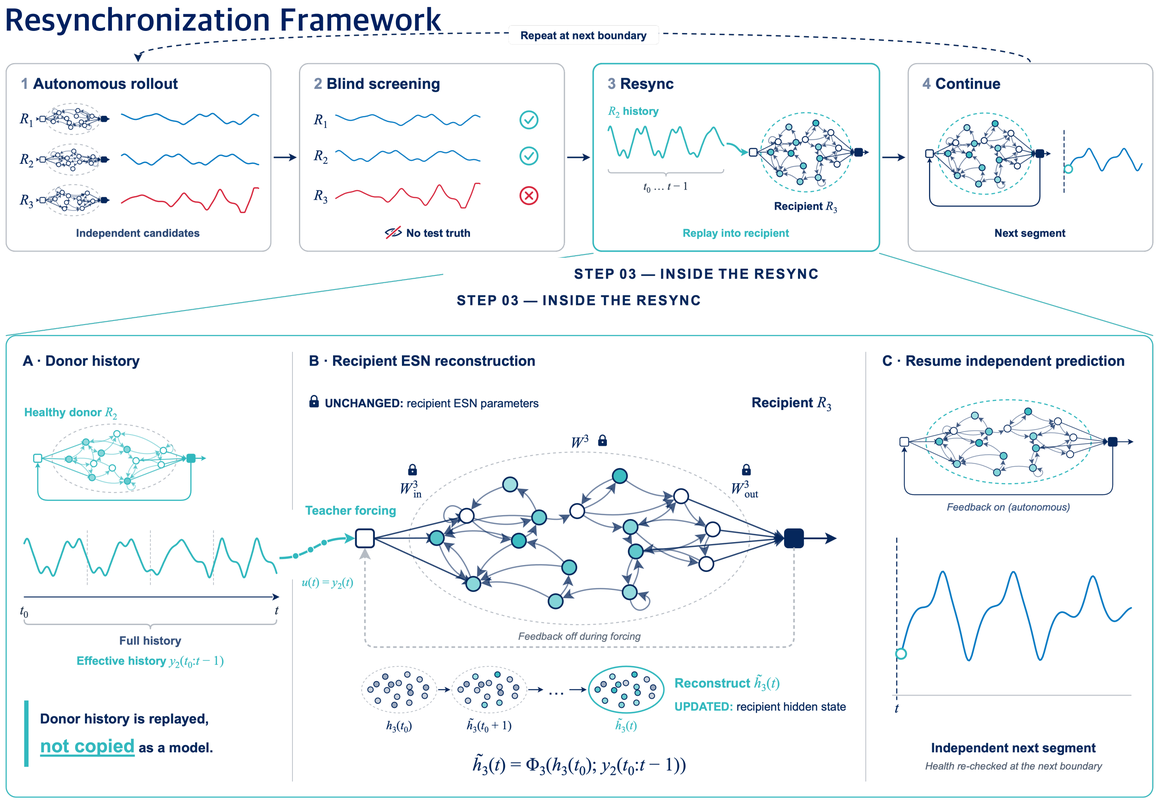
### A bad prediction does not necessarily mean a bad model

We all make mistakes from time to time. Not because we are bad people or inherently flawed—mistakes simply happen. The same is true for ensemble members.

Even when an ensemble is built from well-trained models, periodic checks may reveal that some members are producing clearly unreasonable predictions. But making a mistake does not mean that a model is fundamentally bad, nor does it justify permanently excluding it from the ensemble.

For reservoir ensembles, members that have gone astray can be brought back on track through resynchronization. By repeatedly checking and realigning their states, we can keep the entire ensemble healthier and more diverse, without leaving any member behind.

### Resynchronization Framework




### Implementation Details

**Independent ESNs.** Each ensemble member has its own reservoir and trained readout, while sharing the same training data and hyperparameter settings.

**Blind intervention.** Members are screened using their autonomous candidate trajectories. Screening thresholds are calibrated beforehand using independent reference data; no future observations are used during rollout.

**Full-history reconstruction.** Selected recipients are driven by a donor's effective output history, replayed from the post-training state up to the current segment boundary—not merely the latest segment.

**Parameter preservation.** Each recipient retains its original reservoir and readout parameters. Only its hidden state is reconstructed before autonomous prediction resumes.

## Evaluation

### Effective Healthy Ensemble Size




$$N_{\text{eff}}(b) = \#\text{CC}\left(G_\varepsilon^{(b)}\right)$$

The number of distinct groups among members classified as healthy, after merging near-duplicates.

$G_\varepsilon^{(b)}$: proximity graph of healthy members, with edges where $d_{ij}^{(b)} < \varepsilon$. Reported as $N_{\text{eff}}/N_{\text{eff}}^{\text{truth}}$.

**Healthiness**

1. Healthy-member Fraction

$$f_H(b) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[s_i(b) = \text{healthy}\right]$$

The fraction of members classified as dynamically healthy.

Block-wise hierarchical screening using trajectory-derived dynamical features, with thresholds calibrated on independent reference trajectories (99% level, Bonferroni-corrected).

2. Statistical Fidelity - Cumulative $W_1$

$$W_{1,i}(t) = \frac{1}{D}\sum_{d=1}^{D}\int \left|F_{i,d}^{[0,t]}(u) - G_d^{[0,t]}(u)\right| du$$

$$W_1^{\text{plot}}(t) = \text{median}_i\, W_{1,i}(t)$$

The discrepancy between generated and reference long-term marginal state distributions.

*Common Definition

$$d_{ij}^{(b)} = \frac{1}{w}\sum_{t\in\mathcal{W}_b} \left\|x_i(t) - x_j(t)\right\|_2$$

Window-averaged trajectory distance. The threshold$\varepsilon$ is calibrated from the 5th percentile of independent reference-trajectory nearest-neighbor distances.

**Diversity**

3. Near-duplicate Member Rate

$$r_{\text{dup}}(b) = \frac{1}{N}\#\left\{i \in \mathcal{A}_b : \min_{\substack{j\in\mathcal{A}_b \\ j\neq i}} d_{ij}^{(b)} < \varepsilon\right\}$$

The fraction of all members having at least one excessively similar eligible neighbor.

Eligible members $\mathcal{A}_b$ remain inside the attractor box throughout the window. Denominator: all $N$ members.

4. Ensemble Representativeness - Snapshot JS

$$\text{JS}(p\|q) = \frac{1}{2}\text{KL}(p\|m) + \frac{1}{2}\text{KL}(q\|m)$$

$$m = \frac{p+q}{2}$$

How closely the ensemble snapshot distribution matches the reference attractor distribution.

20×20 projected state-space histograms; lower is better. Reference ensembles provide the null benchmark.

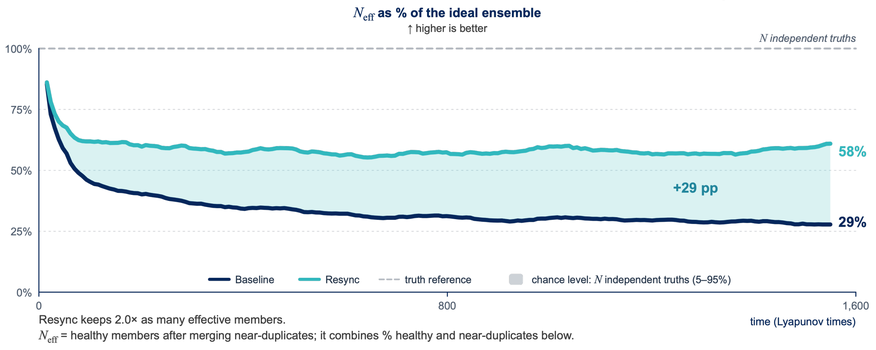
*No future ground-truth observations are used during classification.*

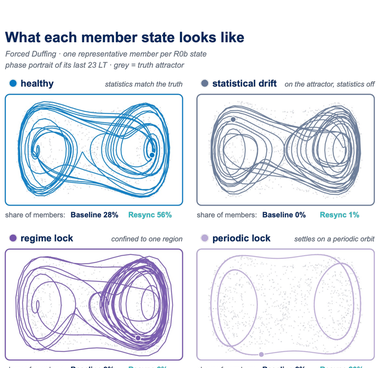
## Results

**Baseline**: Independently trained ESNs run autonomously without intervention. **Resync**: The same autonomous ensemble with blind screening and donor-driven hidden-state reconstruction at segment boundaries. **Truth reference**: Independently generated reference trajectories used for evaluation, not for intervention.

### Forced Duffing (2D)

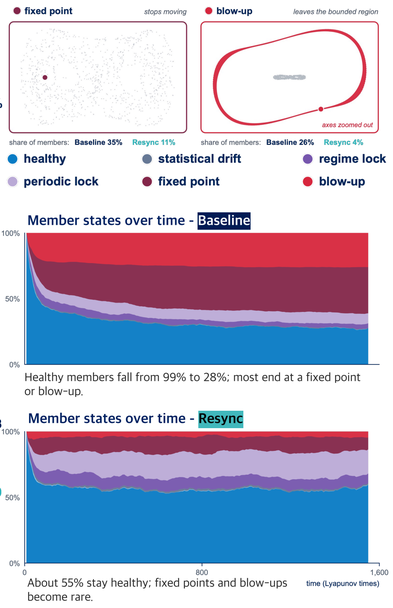
$$\ddot{x} + \delta\dot{x} - \alpha x + \beta x^3 = \gamma\cos(\omega t)$$



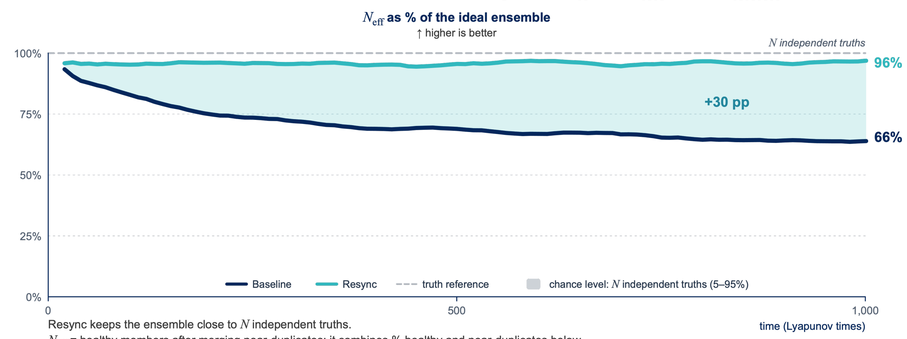


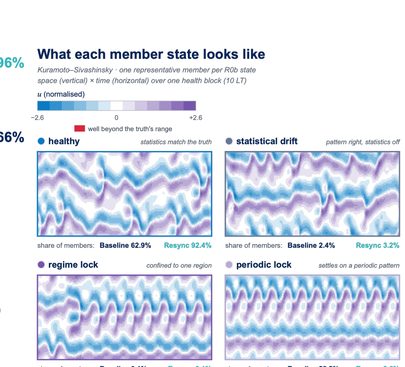


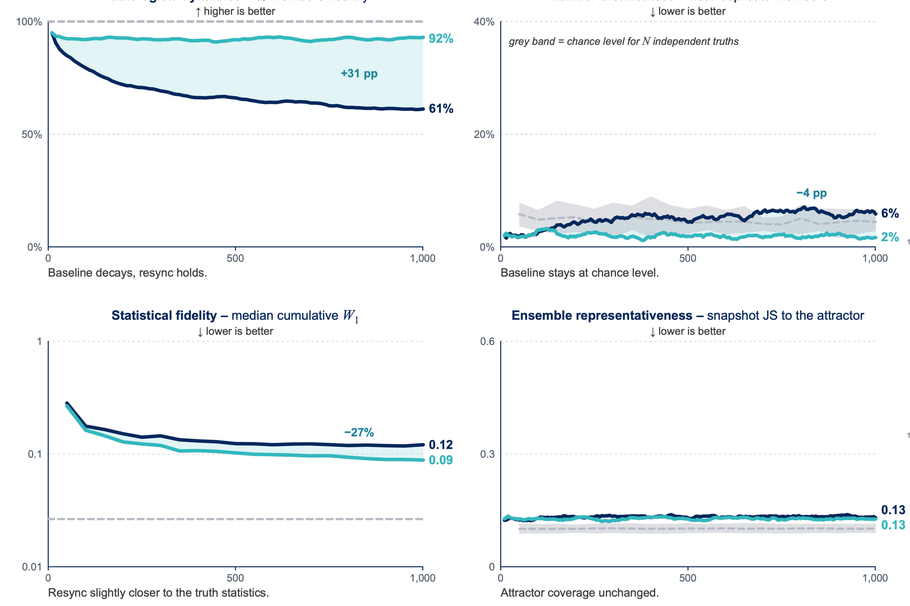


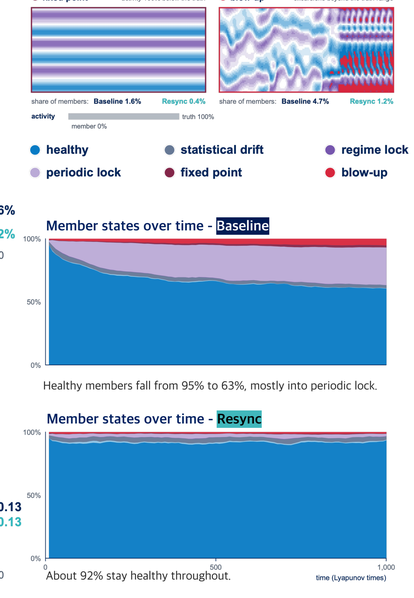

### Kuramoto-Sivashinsky (64D)

$$u_t + uu_x + u_{xx} + \nu u_{xxxx} = 0$$




*Duffing. We use the forced Duffing oscillator $\ddot{x} + \delta\dot{x} - \alpha x + \beta x^3 = \gamma\cos\omega t$ ($\alpha = \beta = 1$, $\delta = 0.3$, $\gamma = 0.4$, $\omega = 1.2$), integrated with RK4 and sampled at $\Delta t = 0.1$ ($\lambda_{\max} = 0.129$; one Lyapunov time ≈ 77 steps). Each ESN has 200 reservoir nodes (spectral radius 0.9, leak rate 0.3, linear readout, ridge $10^{-6}$) and is trained on 5,000 steps. The ensemble of 300 members is forecast for 1,550 LT in segments of $\tau = 7.8$ LT.*

*KS. We use the Kuramoto-Sivashinsky equation u_t + uu_x + u_xx + u_xxxx = 0 on a domain of length L = 22 with 64 grid points, integrated with ETDRK4 and sampled at $\Delta t = 0.25$ ($\lambda_{\max} = 0.042$). Each ESN has 400 reservoir nodes (spectral radius 0.4, leak rate 0.9, bias scaling 0.9, quadratic readout features, ridge $10^{-4}$) and is trained on 20,000 steps. The ensemble of 300 members is forecast for 1,000 LT in segments of $\tau = 5$ LT.*

Resynchronization maintains more dynamically healthy members over long autonomous rollouts, bringing the effective healthy ensemble size closer to the independent-truth reference without increasing near-duplication. However, maintaining healthy members does not necessarily guarantee faithful long-term statistics, highlighting the importance of evaluating statistical fidelity separately.